ENGINEERING PORTFOLIO · FLUIDS

Nathaniel Murcin

Structural and fluid systems design, from concept to hardware.

Georgia Institute of Technology
B.S. Aerospace Engineering, expected May 2027
M.S. Aerospace Engineering, expected May 2028

BLUE ORIGIN · BEN T. ZINN COMBUSTION LAB · GTXR

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Cryogenic tank-farm P&ID design

Blue Origin · Fluids & Manufacturing Engineering Intern · May–Aug 2026 · Cape Canaveral, FL

PROBLEM

- Design a critical-path cryogenic tank-farm P&ID
- Starting point: heritage designs

CRITERIA

- Meet the facility's requirements with the fewest components
- Every line, valve and relief device sized and validated
- Hazards analyzed before formal HAZOP review

DECISION

- Challenged requirements, eliminated branches and downsized hardware
- Sized, selected and validated cryogenic piping, valves and relief devices: Cv, Darcy-Weisbach, line sizing, overpressure relief
- Developed the BOM and coordinated suppliers

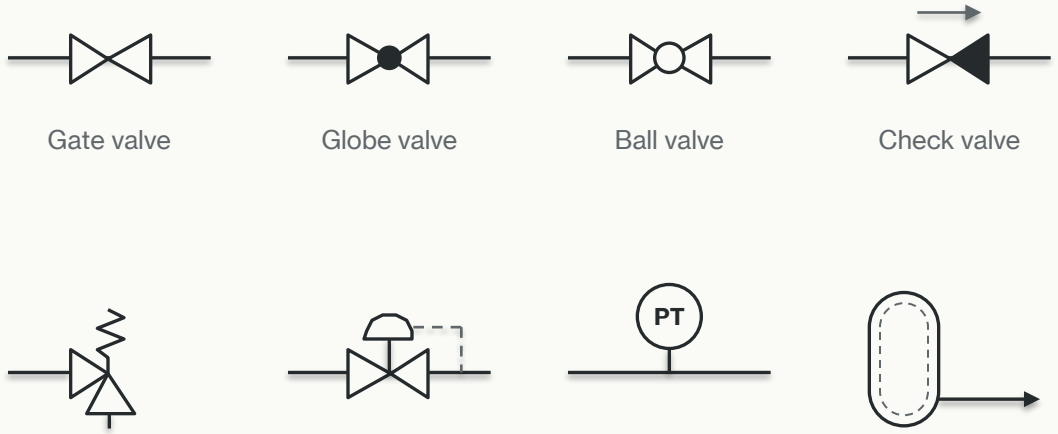
VERIFICATION

- Preliminary hazard analysis: component failure modes and consequences against safeguards
- Documented the CONOPS for formal HAZOP review

~60%

fewer components than heritage designs

P&ID SYMBOLS (GENERIC)



Generic P&ID symbols for illustration, not the Blue Origin system. No Blue Origin hardware, facilities, drawings or specifications are shown.

Multi-gas manifold design

Undergraduate Researcher · Jan 2026–present

PROBLEM

- Reacting jets-in-crossflow experiments need H_2, He, GOx, N_2 and natural gas delivered to predetermined inlets on the main flow rig

CRITERIA

- Regulated delivery of each gas to its assigned inlet
- Hydrogen supply combined from four bottles

DECISION

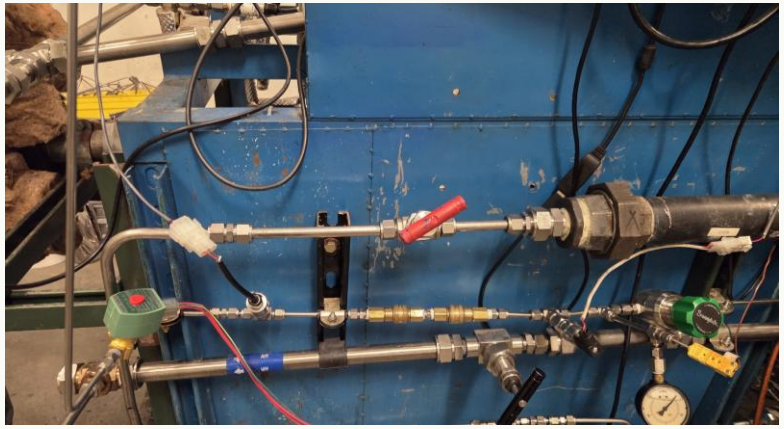
- **Designed and assembled** regulated multi-gas manifold hardware blocks
- Installed the distribution systems, including the piping and manifold tubing that combines four H_2 bottles

OUTCOME

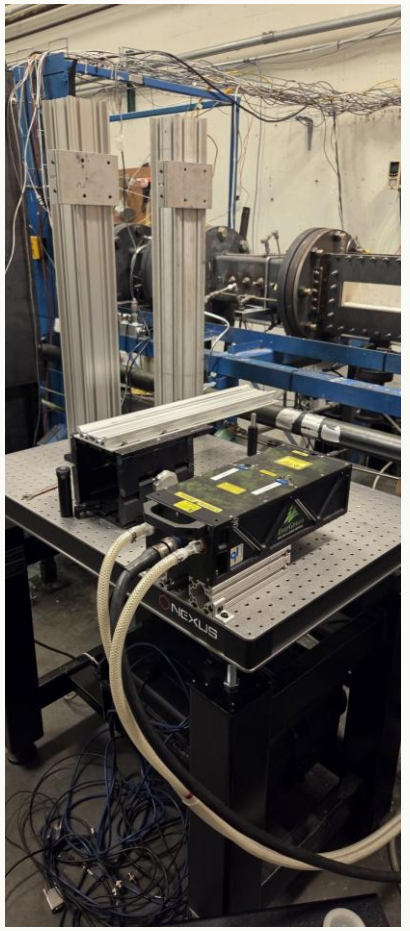
- Systems feed the main flow rig for reacting jets-in-crossflow experiments
- Helped construct the laser PIV setup; post-processed PIV data in MATLAB
- Now building hardware for a follow-on experiment



Tubing combining four H_2 bottles



Gas piping I installed



Laser PIV setup

Flight-critical design and build in 14 hours

Second-stage ignition passthrough, designed and built with the GTXR fin team

PROBLEM

- A flight-unit passthrough failed
- Second-stage ignition needed an electrical path through a pressure boundary

CRITERIA

- Seal at the NPT pressure-boundary interface
- Carry the electrical connection for second-stage ignition
- Pass a hydrostatic proof test at 2× MEOP before flight

DECISION

- Designed and fabricated a replacement flight-critical NPT passthrough with the fin team in **14 hours**

VERIFICATION

- Hydrostatic proof test at 2× MEOP, run by the propulsion lead
- Hardware flew; **the second stage ignited**
- The design informed later passthroughs



Finished NPT passthroughs

14 hr

failure to finished flight hardware

2× MEOP

hydrostatic proof (run by the propulsion lead)

Flew

second stage ignited

Fin design trade study

Fin Design & Manufacturing Lead, Georgia Tech Experimental Rocketry · Aug 2023–present

PROBLEM

- Choose a fin cross-section and core for eight flight fins

CRITERIA

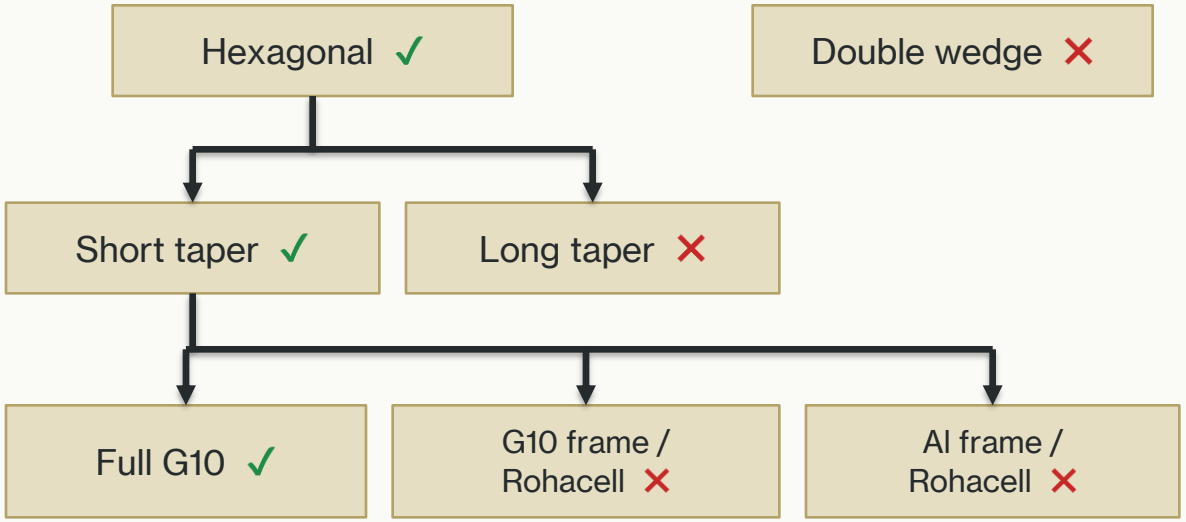
- Manufacturability & repeatability · mass · aerodynamic performance · process safety
- Goal: consistent across eight fins, **not maximum single-fin performance**

DECISION

- Hexagonal over double wedge: repeatable, higher predicted flutter margin; modest aerodynamic penalty accepted
- Short taper over long taper: easier machining, higher flutter margin
- Full G10 core over G10 frame + Rohacell (process not yet mature) and Al frame + Rohacell (CTE mismatch, prior delamination)

VERIFICATION

- Simulation results informed the architecture; the GTXR simulations team later optimized detailed geometry
- Booster recovered with fins intact on both flight vehicles



1.72 lb
Full G10 core (selected)

0.48 lb
G10 frame + Rohacell

+1.24 lb
accepted for repeatability and flutter margin

Precision fin alignment jig

PROBLEM

- Bond four fins at 90° spacing and 1.6° cant, held through cure
- Practice article built with the previous method failed clocking (1.041°)

CRITERIA

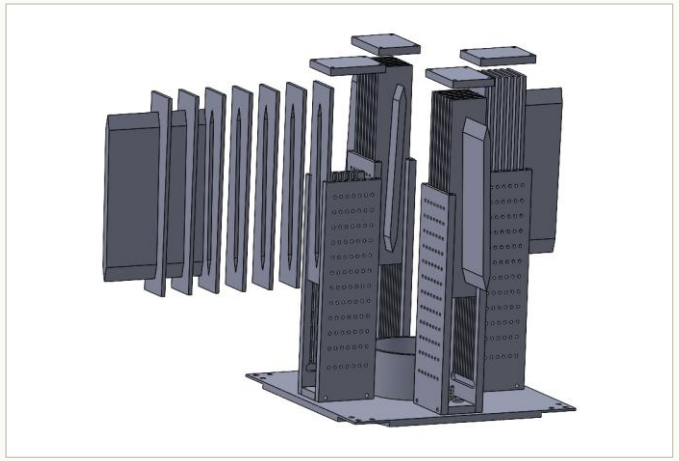
- Clocking 90° ± 1° (vehicle sensitivity analysis)
- Cant 1.6° ± 0.1°
- Adapts to multiple rocket diameters and fin geometries

DECISION

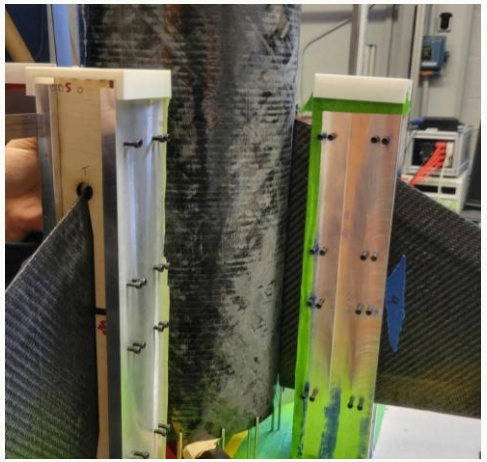
- Led design of a **modular four-tower fixture** (SolidWorks) and its operating procedure; aided CAM and machining
- Iterated the process and usage instructions; standardized layup and attachment

VERIFICATION

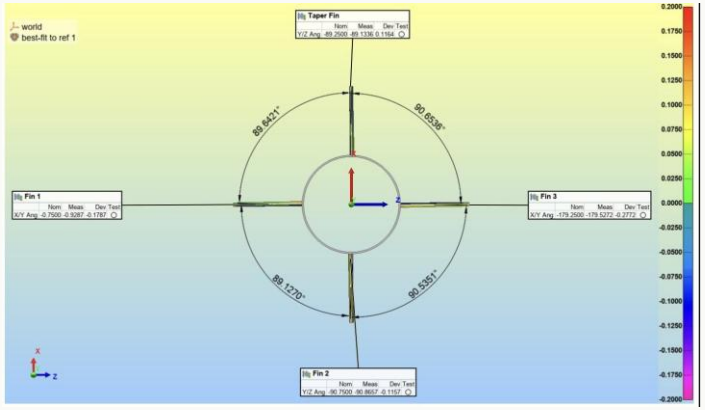
- Practice article, 3D scan: worst-case clocking 1.041° → 0.277° (~73%)
- Flight fins: clocking and cant **within 0.1°**
- ~1.5 hr labor saved; 24 hr cure-time reduction



SolidWorks model



Fixture in use



Practice article re-scan: 0.277° worst case

0.277°

practice-article worst case (from 1.041°)

< 0.1°

flight fins, clocking and cant

Flight fins are not pictured: the scan shown is of the practice article, made before the flight fins. The flight fins were scanned separately; results are from those scans.

Lunar tooling structural analysis

Blue Origin · Fluids & Manufacturing Engineering Intern · May–Aug 2026 · Cape Canaveral, FL

PROBLEM

- Grip lunar hardware without part slippage or damage

CRITERIA

- Positive margins against yield and ultimate for every load case

DECISION

- Analyzed the fixture **like a flight structure**
- Hand calculations for each load case, cross-checked with Creo FEA

VERIFICATION

- Each load case evaluated against the yield and ultimate margin criteria

STRUCTURAL ANALYSES PERFORMED

Torque resistance	Bolt shear & tension
Preload	Joint slip
Tear-out	Bending
Thermal expansion (CTE)	Creo FEA cross-check

No Blue Origin hardware images or specifications are shown.

Nathaniel Murcin

My work centers on mechanical and fluid systems design.

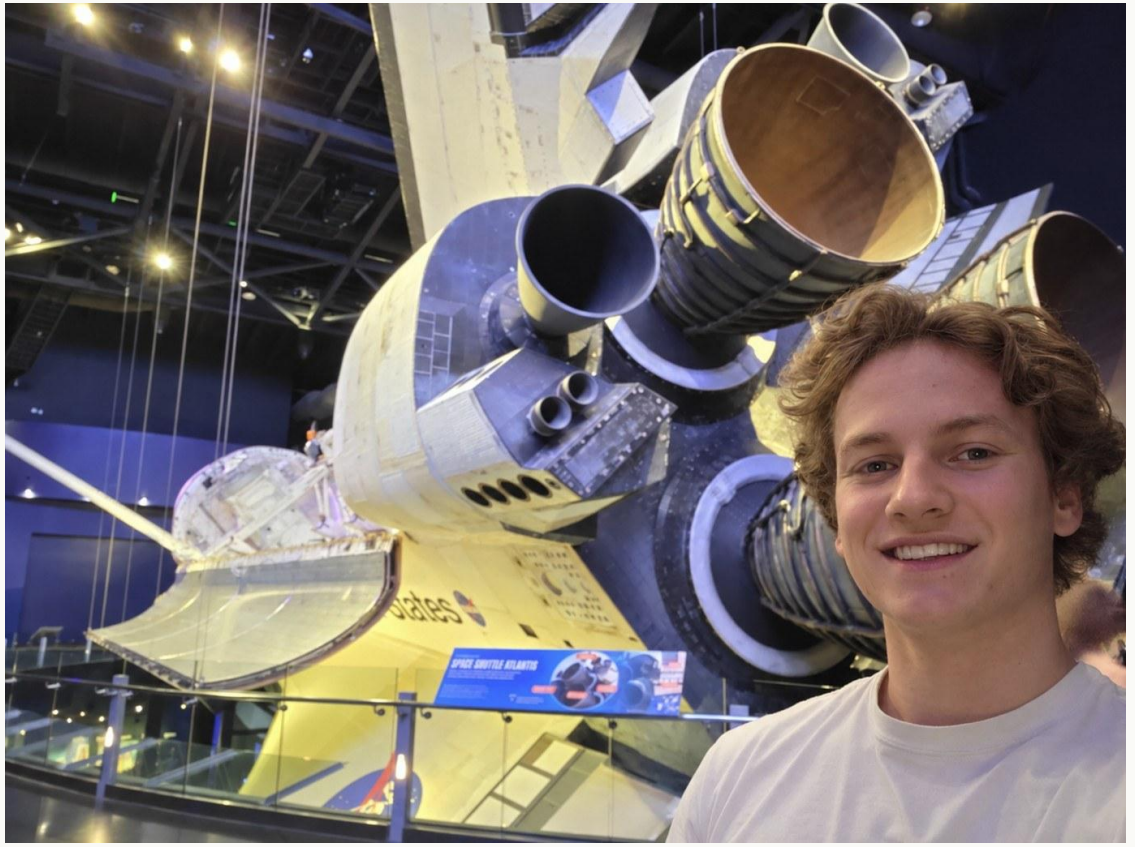
Aerospace Engineering · Georgia Institute of Technology

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Minor in Astrophysics · GPA 4.00/4.00

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Space Shuttle Atlantis, Kennedy Space Center